Java Assignment - 1

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- **Date:** 15-03-2026
- **Topics:** Java Introduction, Compiler, Text Editors vs IDEs, Classes and Packages

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Date: Mar 11, 2026

1. The First Program

HelloWorld.java

```
public class HelloWorld {
    public static void main (String [] args) {
        System.out.println("Hello, world!");
    }
}
```

Expected Output:

```
Hello, world!
```

Can be (compiled and) executed in any text editor or IDE.

The `HelloWorld.java` file's execution in VSCode.

2. Compiler

Translates source code(.java file) into binary bit code (.class file)

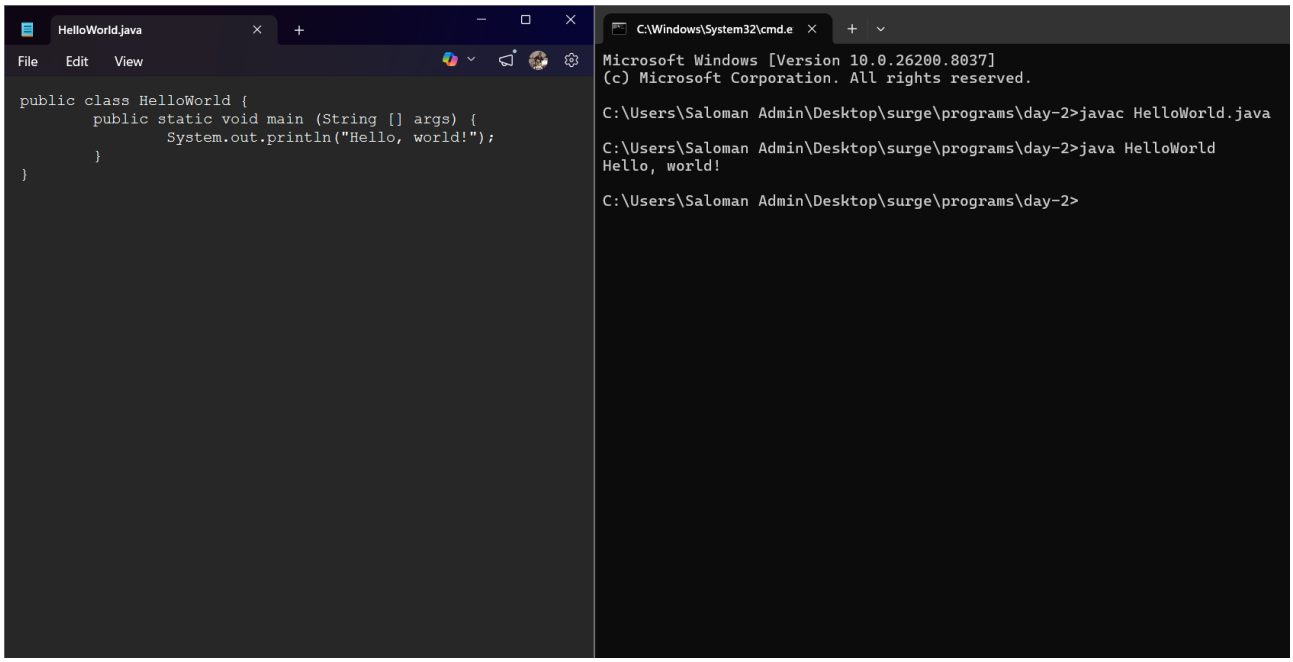
Syntax:

```
> Javac <program_name>.java     //compile
> Java <program_name>           //run
```

Example:

HelloWorld.java —> HelloWorld.class

```
> Javac HelloWorld.java         //compile
> Java HelloWorld               //run
```



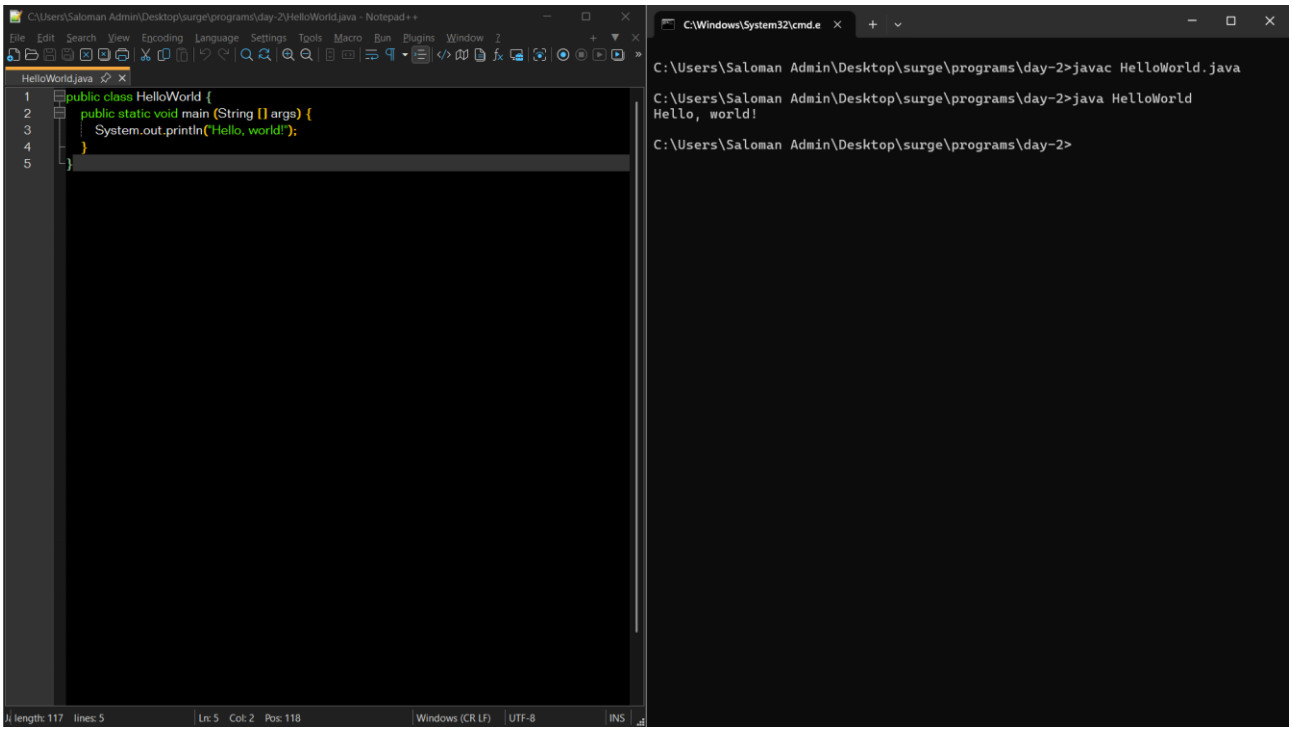
3. Text Editors VS IDEs

Text Editors: Notepad, Notepad++, Sublime Text, etc.
IDEs: Visual Studio Code, Eclipse IDE, IntelliJ IDE, etc.

a) Text Editors

> 1. Write Code.
> 2. Save file.
> 3. Compile manually in command prompt in the respective directory.
> 4. If errors exist, correct them manually in file until there's no errors. And save again.
> 5. Run/execute for Output.

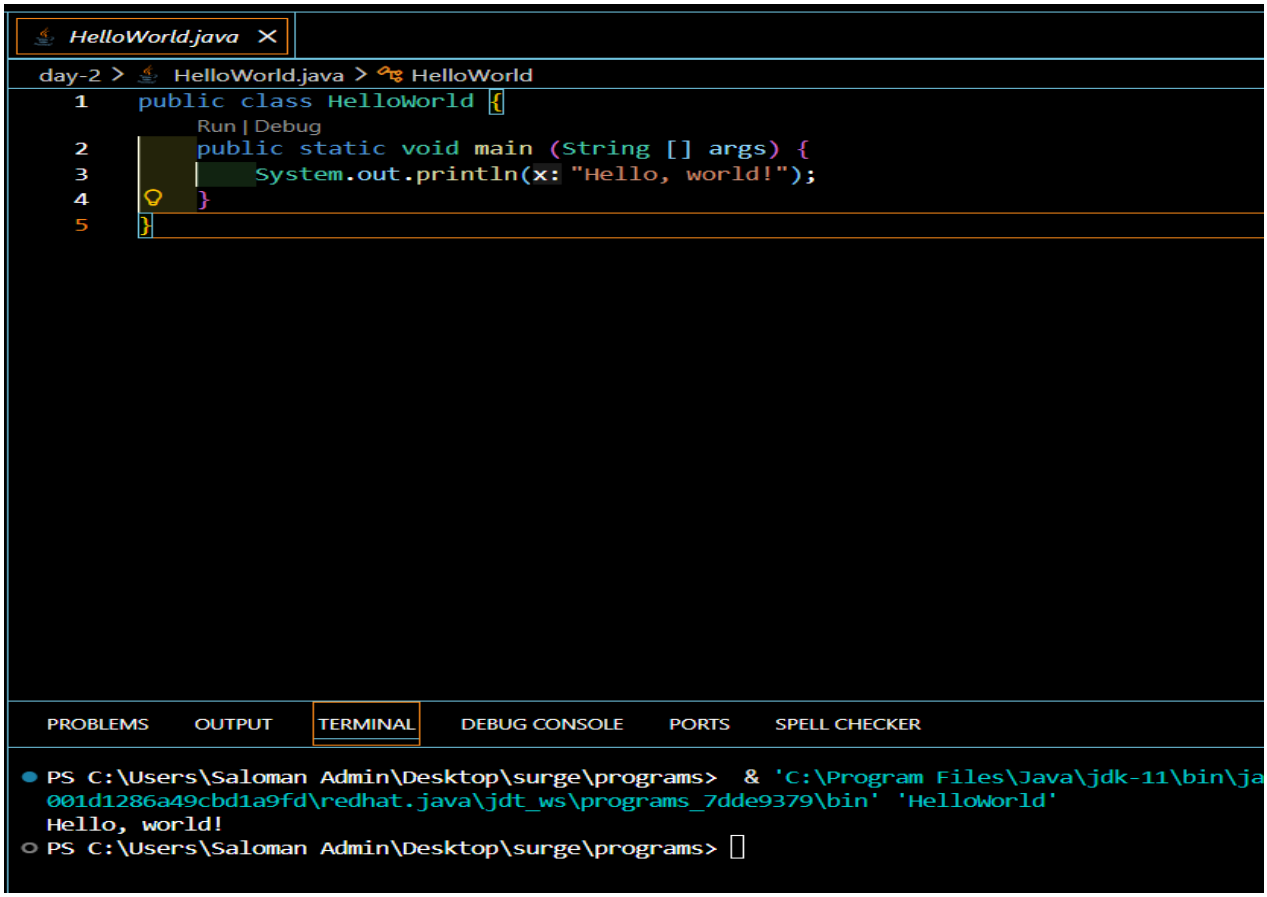
Ex: Notepad++



b) IDEs

> 1. Write code.
> 2. Save file.
> 3. Run/execute directly.

Ex: Visual Studio Code



Date: Mar 12, 2026

4. Class

The template for creating an object.

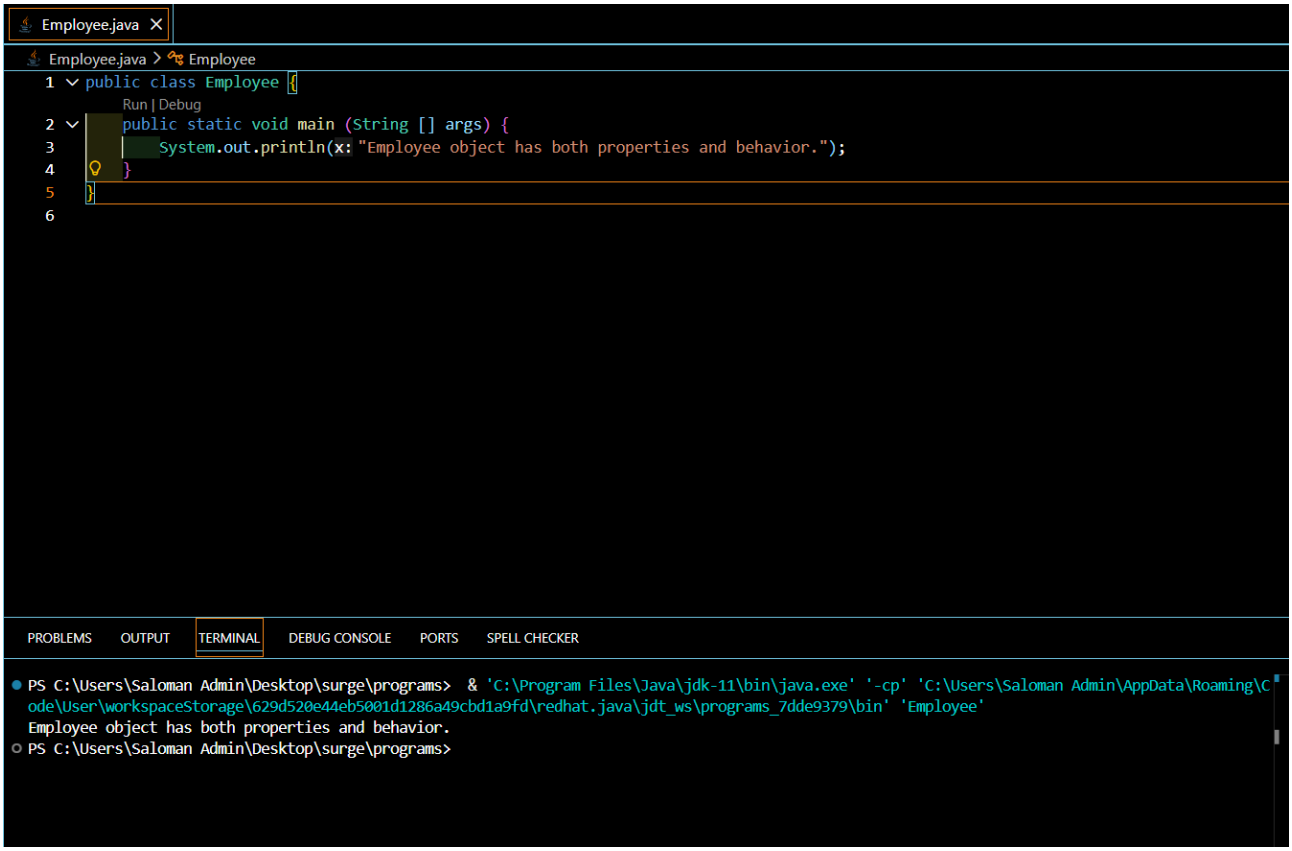
Syntax:

```
public class <class_name> {}
```

Example:

```
public class Employee {}
```

Example in VSCode:



5. Packages

The container/folder which holds the multiple java classes together.

Syntax:

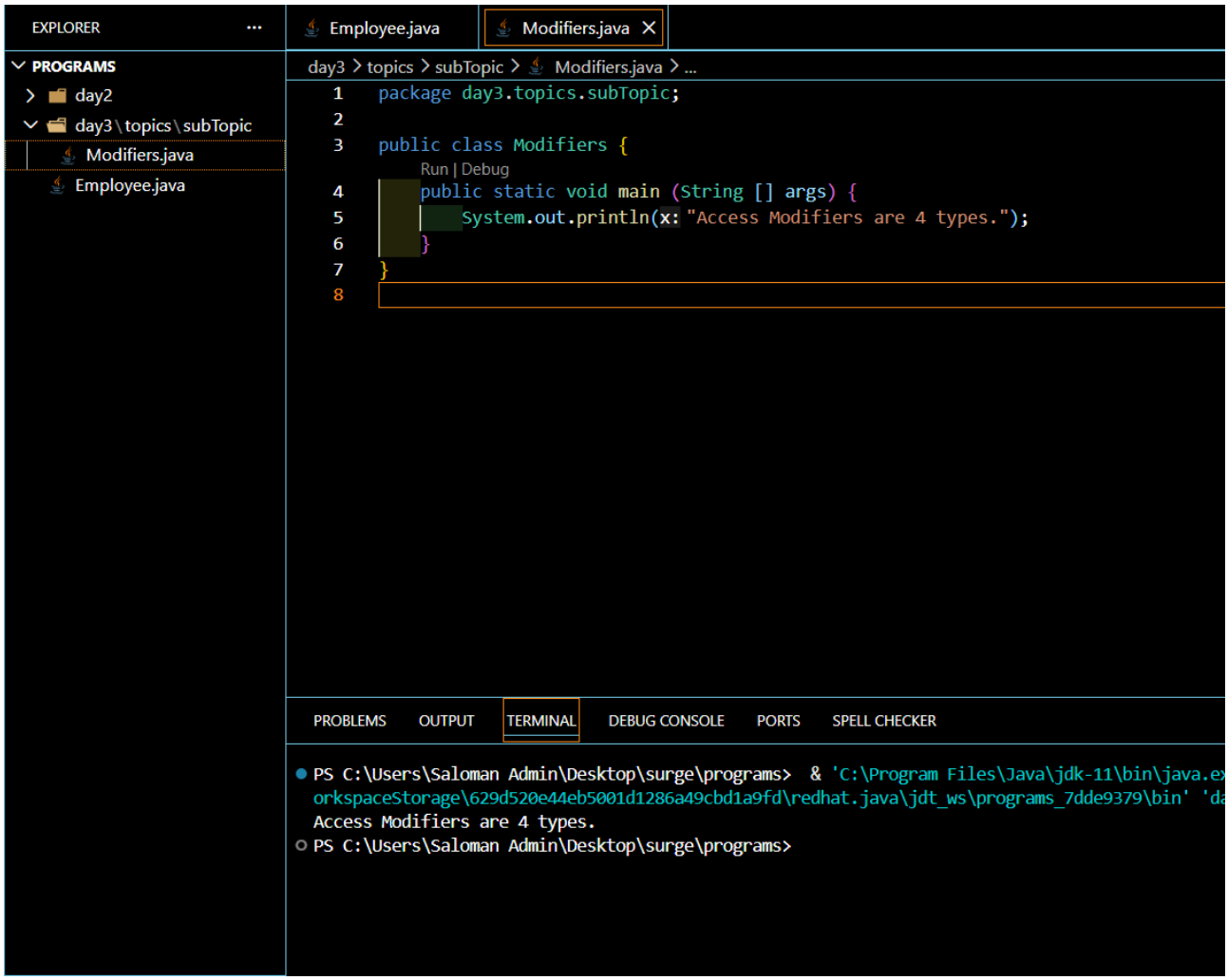
```
package <package1Name>.<package2Name>.<package3Name>...;
```

- Dot (.) is used to connect the packages.

Example:

```
package day3.topics.subTopic;
```

Example in VSCode:



The structure of the packages and files is like the following for above code in `day3` package:

```
day3 (super package of all)
    |
    |- topics (super package of current package)
        |
        |- subTopic (current package)
            |
            |--Modifiers.java (current file)
```